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Title of Research Report

Hybrid Neural Force Correction for the Unrestricted Three-Body Problem

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Hybrid Neural Force Correction for the Unrestricted Three-Body Problem

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Abstract

We present SIMON, a hybrid integrator combining a second-order leapfrog scheme with a learned scalar pairwise force correction and two physical constraints: an analytic force direction that preserves Newtonian radial symmetry, and adaptive sub-stepping that resolves close gravitational encounters. Through a controlled four-run ablation study—removing each component in isolation—we show that long-horizon orbital stability over 100 years depends on these structural constraints: adaptive sub-stepping eliminates the timestep-frontier ejection failure (105× worst-case frontier error reduction), while the learned-direction baseline demonstrates, for the tested initial condition, that replacing analytic direction with a neural approximation produces unbounded trajectories. The neural network correction itself provides an 8.5% accuracy improvement without contributing to stability. Evaluated across seven distinct three-body initial conditions spanning near-circular to tight-binary dynamical regimes, SIMON maintains bounded dynamics for all stable configurations tested (mean divergence rate $\lambda = 0.092 \pm 0.068\ \text{yr}^{-1}$), and agrees with the reference integrator on the occurrence of ejection in the three dynamically unstable configurations; post-ejection separations differ, as expected, because chaotic divergence amplifies integrator differences after the system becomes unbound. Energy drift scales as dt^2, consistent with second-order leapfrog theory, with negligible contribution from the neural correction. At the operational timestep ($dt = 0.04$ yr), SIMON achieves a 1.12× speedup over `ias15` with simulation performance invariant across five independent training seeds. These results demonstrate that long-horizon stability in hybrid neural integrators for chaotic gravitational systems requires structurally enforced physical constraints rather than learned approximations alone.

Keywords: Three-Body Problem; Neural Networks; Hybrid Integrator; Leapfrog Integration; Adaptive Sub-Stepping; Chaotic Dynamics; Gravitational N-Body Simulation; Lyapunov Divergence

Acknowledgement

[Acknowledgement text to be completed by author.]

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3. To the best of our team's knowledge, out paper does not contain any published or written research results by others, other than those listed in the references and acknowledgements. There is no ghostwriting, plagiarism, or other violations; and there are no additional advisors.
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1 Introduction

The Three-Body Problem—the gravitational dynamics of three mutually interacting massive bodies—represents one of the longest-standing challenges in classical mechanics. While exact analytical solutions exist for two-body systems, three-body dynamics are generically chaotic: trajectories are exquisitely sensitive to initial conditions, and long-horizon prediction is fundamentally limited by exponential divergence. Despite this inherent unpredictability, accurately simulating three-body trajectories over timescales of decades to centuries remains essential for a wide range of astrophysical applications, including the characterisation of exoplanet stability zones, stellar cluster dynamics, and spacecraft trajectory design [Gardner et al., 2006].

Traditional step-by-step numerical integrators—such as `ias15`, a 15th-order Gauss-Radau method [Rein and Spiegel, 2014]—achieve near machine-precision energy conservation but scale linearly in computational cost with the simulation horizon. Recent work has proposed hybrid approaches in which neural networks supplement conventional force evaluations [Ulibarrena et al., 2024], offering faster simulation at the cost of some physical fidelity. However, unconstrained neural force models frequently fail to maintain orbital stability during close encounters, where the gravitational force gradient becomes extremely steep and small positional errors are catastrophically amplified [Dehnen, 2001].

This work presents `SIMON`, a hybrid integrator that embeds two complementary physical constraints directly into the neural force evaluation pipeline: (1) an *analytic force direction* that preserves the exact radial symmetry of Newtonian gravity at every step, and (2) an *adaptive sub-stepping mechanism* that resolves close encounters by dynamically reducing the timestep when body separations fall below a threshold. Through a controlled ablation study—removing each constraint in isolation while holding all other parameters fixed—we provide direct experimental evidence for the distinct role of each mechanism, demonstrating that orbital stability over a 100-year horizon requires both constraints working together.

We address the following research question:

> *How closely can physically constrained neural network force models approximate the long-term stability of traditional numerical integrators in three-body simulations, while reducing computational cost without sacrificing qualitative dynamical stability?*

The key contributions of this work are:

1. A controlled ablation study isolating the contributions of adaptive sub-stepping, analytic force direction, and neural network correction to long-horizon orbital sta-

bility.

2. Generalisation evaluation across seven distinct three-body initial conditions, spanning near-circular, hierarchical, moderate scattering, and tight-binary dynamical regimes.

3. Quantitative confirmation that `SIMON`'s energy drift scales as dt^2, consistent with second-order leapfrog theory, with the drift attributable to the integrator rather than the neural correction.

4. A training seed robustness analysis confirming that simulation performance is invariant to random weight initialisation.

5. A speed-accuracy frontier demonstrating $1.12\times$ speedup over `ias15` at the operational timestep, with bounded dynamics maintained across all tested stable configurations.

The paper is structured as follows. Section 2 reviews the three-body problem, chaos, and relevant neural approaches. Section 3 discusses computational modelling challenges. Section 4 describes the `SIMON` architecture and experimental setup. Section 5 presents experimental results across four themes: stability decomposition, generalisation, physical consistency, and computational performance. Section 6 discusses implications, limitations, and future directions.

2 Literature Review

The literature relevant to this project lies at the intersection of celestial mechanics, numerical integration, and scientific machine learning. Classical mechanics explains why the unrestricted three-body problem is sensitive to initial conditions and difficult to solve analytically, while numerical-analysis literature shows that long-horizon behaviour depends strongly on whether an algorithm preserves geometric structure and handles close encounters robustly Hairer et al. [2006], Rein and Tamayo [2015], Rein and Spiegel [2014], Boekholt and Portegies Zwart [2015]. More recent machine-learning work shows that neural networks can act as learned simulators or learned Hamiltonian surrogates, but also that the success of these models depends on the particular inductive biases built into the architecture Chen et al. [2018], Battaglia et al. [2016], Greydanus et al. [2019], Gruver et al. [2022], Horn et al. [2025]. This review therefore moves from classical gravitational dynamics to modern learned simulators and ends by identifying the specific methodological gap addressed by `SIMON`.

The Three-Body Problem is the classical problem in celestial mechanics of describing the motion of three massive bodies gravitationally interacting with each other. This problem has significant applications in understanding the orbits of planets, moons, and artificial satellites [Gardner et al., 2006].

The Three-Body Problem is the manifestation of centuries of development in astronomy. Its history dates back to ancient Greek times. The Greeks were among the first to begin the systematic observation of celestial movements, although they lacked the mathematical framework to describe the complex gravitational interactions between bodies precisely [Anjum and Mishra, 2020]. However, in the 17th century, Sir Isaac Newton's *Philosophiæ Naturalis Principia Mathematica* [Newton, 1687, Cohen and Whitman, 1999] dictated universal laws that laid the groundwork for classical mechanics, creating a wide range of avenues for research into gravitational systems. To utilize Newton's formalism to understand the Three-Body Problem, we first need to analyze the simpler Two-Body Problem.

2.1 The Two-Body Problem

The general formula for the force of attraction between any two objects in a gravitational field is given by Newton's Universal Law of Gravitation in Equation 1:

$$F = G\frac{m_1 m_2}{r^2}, \tag{1}$$

where G is the universal gravitational constant, m_1 is the mass of the first body, m_2 is the mass of the second body and r is the distance between the centers of the bodies. Using this formula, the total force of gravity experienced by an object from any number of massive bodies can be calculated.

To understand how this force translates to celestial orbital motion, consider the following scenario: an observer on earth throws a ball of mass m_b. The harder the ball is thrown, the farther it will travel, but will eventually fall to the ground as a result of gravitational attraction exerted on it by the earth. However, if thrown hard enough, the ball will make an orbit around the earth and return to where it started. If we assume that the ball moves in a circular orbit around the Earth, we can equate the force of gravity exerted on the ball by the earth to the centripetal force on the ball, giving us Equations 2, and 3.

$$G\frac{m_e m_b}{r^2} = \frac{m_b v^2}{r} \tag{2}$$

$$v = \sqrt{\frac{Gm_e}{r}}, \tag{3}$$

where m_e is the mass of the Earth, v is the velocity of the ball, G is the universal gravitational constant, and r is the distance between the two bodies, measured from their centers of mass.

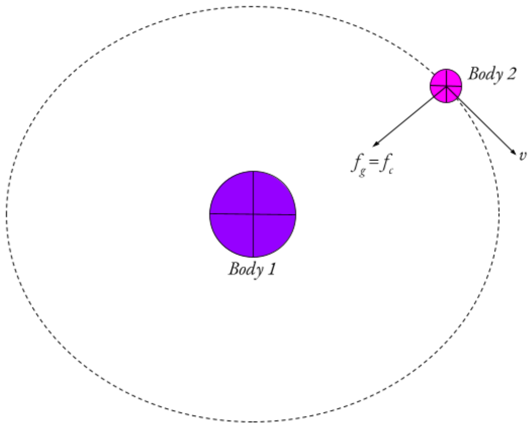

Figure 1: A diagram of a simple two-body system.

Similarly, a two-body system (see Figure 1) can be reduced to operate under the same principles as the ball and the earth example. Therefore, the velocity of the orbiting planet can be given by Equation 3. In the math presented above, the path of the ball around the Earth is approximated to be circular. However, in reality, the ball can take any elliptical path around the Earth [Murray and Dermott, 1999]. The two-body framework can be used to model systems such as a satellite orbiting the earth or a planet orbiting the sun. In addition, the two-body problem is helpful in modeling N-body systems where the masses of two objects are much larger than the masses of all other bodies. However, the two-body problem omits situations in which three or more masses are all exhibiting comparable gravitational forces on each other at the same time.

2.2 The Three-Body Problem

In the three body problem, the goal is to approximate the future positions and velocities of the *three* planets, given their initial states. However, when calculating the future states of a system of three bodies interacting gravitationally, it is not possible to construct an equation that will precisely reveal the trajectories of each body. To obtain this conclusion, we define some assumptions and simplifications.

- **Point Masses:** The bodies will be treated as point masses.
- **Newtonian Gravity:** We will use Newton's Law of Gravitation to describe the force of gravity.
- **Isolated System:** The system only consists of the three bodies interacting, without external forces.

We present a schematic of this system in Figure 2.

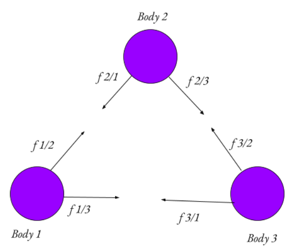

Figure 2: A diagram of a simple three-body system. Each body is exerting a gravitational force on the other two bodies.

We will use a three-dimensional cartesian coordinate system to define the centers of each of the bodies. Let the position vector of a body n be the vector $\mathbf{P}_n$. Each of the six forces in the system can be described using Newton's law of gravitation, generally given by

$$\mathbf{F}_{n/b} = G\frac{M_n M_b}{|\mathbf{P}_n - \mathbf{P}_b|^3}(\mathbf{P}_n - \mathbf{P}_b)$$

Where $\mathbf{F}_{n/b}$ is the gravitational force of attraction that body n feels toward body b. This is the first law we will use to formulate our problem. We will also take Newton's second law as follows.

$$\sum \mathbf{F}_n = m\mathbf{a_n}$$

$$\sum \mathbf{F}_n = m\left(\frac{d\mathbf{v}_n}{dt}\right)$$

$$\sum \mathbf{F}_n = m\left(\frac{d^2\mathbf{P}_n}{dt^2}\right)$$

We can now use these two laws to create a system of Ordinary Differential Equations (ODEs) that we can use to approximate the positions of the bodies at time t. We can take the sum of forces on body 1 as an example:

$$\sum \mathbf{F}_{\text{exterior}}(\text{body 1}) = m_1\frac{d^2\mathbf{P}_1}{dt^2} = \mathbf{F}_{2/1} + \mathbf{F}_{3/1}$$

Substituting in law 1:

$$\sum \mathbf{F}_{\text{exterior}}(\text{body 1}) = G\left[M_2\frac{\mathbf{P}_2 - \mathbf{P}_1}{|\mathbf{P}_2 - \mathbf{P}_1|^3} + M_3\frac{\mathbf{P}_3 - \mathbf{P}_1}{|\mathbf{P}_3 - \mathbf{P}_1|^3}\right]$$

$$\frac{d^2\mathbf{P}_1}{dt^2} = G\left[M_2\frac{\mathbf{P}_2-\mathbf{P}_1}{|\mathbf{P}_2-\mathbf{P}_1|^3} + M_3\frac{\mathbf{P}_3-\mathbf{P}_1}{|\mathbf{P}_3-\mathbf{P}_1|^3}\right]$$

Doing the same for bodies 2 and 3 we get the system of ODEs [Murray and Dermott, 1999]:

$$\frac{d^2\mathbf{P}_1}{dt^2} = G\left[M_2\frac{\mathbf{P}_2-\mathbf{P}_1}{|\mathbf{P}_2-\mathbf{P}_1|^3} + M_3\frac{\mathbf{P}_3-\mathbf{P}_1}{|\mathbf{P}_3-\mathbf{P}_1|^3}\right]$$

$$\frac{d^2\mathbf{P}_2}{dt^2} = G\left[M_1\frac{\mathbf{P}_1-\mathbf{P}_2}{|\mathbf{P}_1-\mathbf{P}_2|^3} + M_3\frac{\mathbf{P}_3-\mathbf{P}_2}{|\mathbf{P}_3-\mathbf{P}_2|^3}\right]$$

$$\frac{d^2\mathbf{P}_3}{dt^2} = G\left[M_1\frac{\mathbf{P}_1-\mathbf{P}_3}{|\mathbf{P}_1-\mathbf{P}_3|^3} + M_2\frac{\mathbf{P}_2-\mathbf{P}_3}{|\mathbf{P}_2-\mathbf{P}_3|^3}\right]$$

This system of ODEs does not have a closed-form analytical solution like the two-body example in Section 2.1. This is because the motion of the three bodies that all exert forces simultaneously on each other is too complex and generally leads to chaos. In this context, chaos means that even the slightest change in position or velocity has a relatively huge impact on the long-term motion of the system, leading to unpredictable motion which makes it impossible to measure with complete accuracy.

2.3 Chaos

Chaos can be quantified using many different frameworks. For the sake of simplicity, the framework we will be using is the Lyapunov Exponent [Ott, 2002] which measures the rate of trajectory divergence as an exponential model. If the initial separation between two points in phase space is $\delta(0)$, their separation at time t is approximated by the function

$$\delta(t) \approx e^{\lambda t}\delta(0)$$

where λ is the Lyapunov exponent. If $\lambda > 0$, the system is chaotic, and if $\lambda \leq 0$ it is regular, or periodic. λ is a time constant.

The ideas of chaos in the Three-Body Problem can be traced back to the late 19th century, when Henri Poincaré introduced the groundwork for defining Chaos [Poincaré, 1890]. He found that when perturbations (deviance due to an external force) in a chaotic system occurred, the stable and unstable manifolds (two different sets of future states of a system, one stable versus unstable) of a periodic orbit began to intersect. Instead of one manifold neatly returning to its source, they crossed each other an infinite number of times. Because they cannot cross themselves but must cross each other to satisfy the equations of motion, they create an incredibly intricate, mesh-like structure. This tangling, known as the homoclinic tangle, leads to chaos because it creates an infinite, fractal-like web of trajectories, that makes the system's future state impossible to analytically predict.

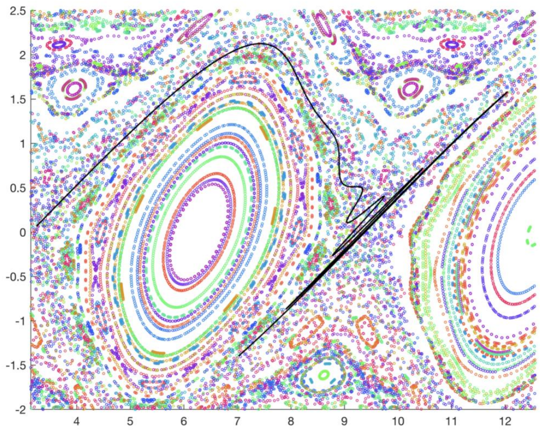

Figure 3: A plot of a homoclinic tangle.

Poincaré himself noted he couldn't even begin to draw the figure, concluding that an algebraic formula to predict a planet's long-term position was physically impossible. Linking back to the Lyapunov exponent, Poincaré predicted that the rate of trajectory divergence would be exponential. Two bodies starting at almost identical positions within the tangle will be stretched along the unstable manifold and compressed along the stable one. Very quickly, their paths smear exponentially across the web.

This work showcased that the Three-Body problem is highly sensitive to its initial conditions, exemplifying how seemingly simple gravitational systems can exhibit unpredictable behavior. This would go on to establish the idea that while some solutions existed, a generalized solution to the Three-Body Problem was impossible, which set up the connection between chaos theory and celestial mechanics.

Some specific factors that influence the chaos of a Three-Body system are mass, relative velocity, and initial separation. For example, if the mass of a system is concentrated on one body while the other two are smaller, then the system is less likely to be as chaotic than if the masses were distributed Hietarinta and Mikkola [1993]. Furthermore, if the relative velocities of the bodies are high, even when nearby, the bodies may be moving with too much kinetic energy to induce an exchange that would cause them to diverge much from their initial trajectories. This would reduce the rate at which changes in trajectory would happen, lowering λ. Lastly, if the bodies have a large initial separation, it is unlikely that they will enter each other's zones of gravitational influence, so the likelihood of divergence from their trajectories will be lower, reducing λ and hence the chaos of the system.

2.4 The Restricted Three-Body Problem

There exists another form of the Three-Body Problem by the name of the Restricted Three-Body Problem, which was developed following Newton's discoveries primarily by Euler and Lagrange during the 18th century. Euler built upon the first versions of the restricted Three-Body problem by fixing two massive bodies in space and solving analytically for the motion of a third, smaller particle under their combined gravitational pull Bistafa [2021]. This makes the problem easier to interpret, with one notable method of interpretation being Lagrange points. Developed by Joseph-Louis Lagrange, these are five special solutions which represent stable positions where a smaller body can theoretically maintain its position Lagrange [1772]. However, these solutions tended to be unstable in the context of the Unrestricted Three-Body problem.

The fact that these stable Lagrange points can be identified means that the restricted Three-Body Problem tends to be more applicable than the more chaotic Unrestricted version, because the restrictions simplify the system enough to be studied to a great degree of accuracy without losing too much of the complex dynamics. For example, the restricted Three-Body Problem is used to chart spacecraft trajectories when entering the gravitational influence of celestial bodies. Moreover, identifying Lagrange points in a restricted Three-Body System allows telescopes like the James Webb telescope [Gardner et al., 2006] to stay in an orbit to perform long-term observational missions.

2.5 Expansion into Modern Physics

Progressing into the 20th century, the Three-Body problem continued to attract significant attention. The development of new numerical methods brought new perspectives on chaos and stability in the Three-Body problem. Furthermore, the development of modern computers allowed for numerical simulations of the Three-Body problem via integration, allowing easier investigation into its chaotic nature for longer time periods.

In modern astrophysics, the Three-Body problem finds application in contexts like calculating orbits of planets in multi-body systems, examining interactions in exoplanet systems, and understanding stability in star clusters. On a more recent note, as this paper will outline, machine learning techniques are being used to deepen traditional methodologies of Three-Body analysis and prediction. One such example of a machine learning application was by Ulibarrena et al. [2024]. In their setup, they have created and evaluated a Hamiltonian Neural Network (HNN,) and their findings were that an HNN was able to successfully emulate a traditional step-by-step integrator, at a considerably reduced computational cost. A Hamiltonian Neural Network is a type of NN that composes forces in a way such that energy conservation is structurally preserved. However, this increased complexity was reflected in their results, as the training process was significantly more advanced than that of other methods of Three-Body computation. The Three-Body Problem continues to be an area of interest for modern computational physics problems, and its chaotic nature makes it especially beneficial as a testbed for numerical stability in physics applied machine learning.

2.6 Summary

To summarise, we have defined:

- **The Two-Body Problem:** Assuming circular orbits, using Newton's universal law of gravitation.
- **The Three-Body Problem:** The setup and derivation of a system of ODEs for approximating the motion of an unrestricted Three-Body system.
- **Chaotic Nature of a Three-Body System:** Introduced the idea of Chaos using the Lyapunov exponent (rate of trajectory divergence), and established that the trajectories in a Three-Body system are highly sensitive to the initial conditions, rendering it impossible to predict without approximation. Explored Poincaré's discoveries on homoclinic tangles and their implications on chaos in the Three-Body Problem.
- **Examples of Influences of Initial Conditions on the Chaos of a Three-Body System:** Explained that a system with more distributed masses, closer bodies, and lower relative velocities is likely to be highly chaotic.
- **The Restricted Three-Body Problem:** Analysis of Lagrange points, and how they are applied to real-life scenarios.

3 Computational Modeling of the Three-Body Problem

The gravitational Three-Body Problem can be simulated using computer programs, as discussed in Section 2.5. Because the system of ODEs defining the unrestricted Three-Body Problem does not have a general closed-form solution, its evolution is usually computed incrementally by numerical integration. A traditional integrator advances the state step by step, so long integrations require many force evaluations, and close encounters can amplify small numerical errors into large trajectory differences. High-accuracy tools such as `REBOUND` and `ias15` are therefore essential benchmarks for this project, because they show what reliable long-horizon gravitational integration should look like Rein and Liu [2012], Rein and Spiegel [2014]. At the same time, the numerical-analysis literature shows that long-time performance is not determined only by local accuracy: preserving geometric structure and resolving close interactions are central to physical reliability Hairer et al. [2006], Boekholt and Portegies Zwart [2015].

Neural networks offer a possible route to reducing this computational burden. In some learned-surrogate approaches, a network is trained on many trajectories and used to map from an initial state directly to a future state, thereby skipping some or all intermediate integration steps. In principle, such a direct predictor could reduce the number of calculations needed for long-horizon prediction. In this project, however, `SIMON` does not use a direct state-to-future-state map. It remains a timestep-based hybrid integrator, but uses a neural network only as a local force-correction mechanism during close encounters. This distinction is important: the aim is not to replace numerical integration entirely, but to test whether a learned component can reduce computational cost while preserving the physical structure needed for stable long-horizon rollouts.

A standard neural force model may fail if it does not respect the underlying physics of the Three-Body Problem. This difficulty is most severe during close encounters. According to Newton's law of universal gravitation, the gravitational force between two bodies separated by distance r is proportional to $1/r^2$. Therefore, as r becomes small, the force magnitude and its gradient change extremely rapidly. Even a small learned error in position, force magnitude, or force direction can then produce a large artificial impulse, causing a body to be ejected from the system or to gain non-physical energy Dehnen [2001]. The central modelling question is therefore not simply whether a neural network can approximate a three-body trajectory, but which parts of the Newtonian force law must remain exact for a learned integrator to remain dynamically usable.

This motivates the hybrid approach studied here. `SIMON` keeps the Newtonian radial force direction analytic, uses adaptive sub-stepping to resolve close encounters, and restricts the neural network to a scalar pairwise force correction. The following section situates this design relative to classical integrators, learned simulators, and structure-preserving neural models.

3.1 Learned Simulators, Structure Preservation, and the Research Gap

Classical numerical literature provides the benchmark against which any learned approach must be judged. Geometric numerical integration shows that symplectic and symmetric methods exhibit improved long-time behaviour on Hamiltonian systems because they preserve structure rather than merely reducing local truncation error Hairer et al. [2006]. In astrophysical practice, `WHFast` emphasises fast symplectic long-term orbit integration, while `IAS15` provides a high-order adaptive Gauß–Radau method that handles close encounters and keeps systematic errors below machine precision over extremely long integrations Rein and Tamayo [2015], Rein and Spiegel [2014]. At the highest-accuracy end, arbitrary-precision work with `Brutus` shows that converged solutions in chaotic three-body scattering can require much more than standard double precision and that conventional integrations often diverge from the converged trajectory on an individual-realisation basis Boekholt and Portegies Zwart [2015]. These studies show why close encounters and long-horizon trajectory agreement are intrinsically hard tests.

A second strand of literature asks whether neural networks can replace or accelerate numerical solvers. Neural ODEs represent dynamics by learning the derivative field and integrating it with a black-box solver Chen et al. [2018]. Interaction Networks and graph-network simulators learn multi-object interactions and can roll out dynamical systems for many timesteps, including particle-based systems Battaglia et al. [2016], Sanchez-Gonzalez et al. [2020]. In celestial mechanics specifically, Breen *et al.* showed that a deep neural network trained on arbitrarily precise three-body solutions can approximate a bounded-time planar three-body map at fixed cost and very high reported speedup Breen et al. [2020]. More broadly, physics-informed neural networks represent a separate branch of scientific machine learning in which governing equations are imposed through the training objective rather than through the time-stepping map itself Cuomo et al. [2022]; a recent three-body PINN preprint argues that such equation-informed training can improve prediction quality and cost relative to earlier machine-learning approaches Pereira et al. [2025]. However, these methods primarily learn the evolution law itself; they do not isolate which parts of Newtonian structure must remain exact for long-horizon boundedness in close-encounter regimes.

A third strand introduces physics directly into the learning architecture. Hamiltonian Neural Networks and Lagrangian Neural Networks encode energy-based formulations so that the learned vector field respects a mechanical structure Greydanus et al. [2019], Cranmer et al. [2020]. `SymODEN` extends this idea to systems with control Zhong et al. [2020], while `SympNets` learn symplectic maps directly and show strong generalisation even on Hamiltonian systems as difficult as the three-body problem Jin et al. [2020]. Other work combines learned Hamiltonian structure with explicit symplectic integration; for example, Taylor-nets use symmetric Taylor expansions together with a fourth-order symplectic integrator for long-term prediction Tong et al. [2021]. Later analysis shows that using a symplectic integrator during training can materially improve Hamiltonian neural networks David and Méhats [2023], and recent matched-cost comparisons find that structure-preserving Hamiltonian networks generalise better than physics-unaware multilayer perceptrons, including on a gravitational three-body benchmark Horn et al. [2025]. At the same time, not every physics-inspired inductive bias is equally important: Gruver *et al.* show that some benefits commonly attributed to Hamiltonian neural networks arise from second-order inductive bias rather than from symplecticity or energy conservation alone Gruver et al. [2022]. This makes it scientifically meaningful to ask which constraints matter most in a given physical domain, rather than simply adding more "physics" in an undifferentiated way.

This question is especially important for chaotic systems. Because nearby trajectories separate exponentially, exact pointwise agreement over long horizons is often unattainable even when the qualitative dynamics are correct Margazoglou and Magri [2023]. Recent work on stabilised neural ODEs makes the same point from the machine-learning side: for chaotic dynamics, a useful learned model should remain on the correct attractor or dynamical regime, rather than only minimising short-horizon rollout error Linot et al. [2022]. In the three-body context, the practical goal is therefore not perfect trajectory overlap at arbitrarily long times, but preservation of the correct qualitative behaviour: bounded motion should remain bounded, unstable systems should eject, and numerical energy drift should stay physically interpretable.

The remaining gap is therefore narrower, and more specific, than whether neural networks can simulate physical systems at all. Prior work shows that learned simulators can be fast, that Hamiltonian or symplectic inductive biases can improve extrapolation, and that physics-informed losses can encode governing equations. However, these approaches do not isolate a practical design question that is central to chaotic gravitational motion: when the force law is known, which components should be learned and which should remain analytically fixed? In particular, the literature does not clearly test whether a neural model should learn the full gravitational force vector, or whether Newtonian radial direction should be enforced exactly while learning is restricted to a lower-dimensional scalar correction. It also does not usually separate the stabilising effect of close-encounter timestep refinement from the effect of the neural model itself. `SIMON` is designed around this gap. It combines a leapfrog backbone with analytically enforced radial force direction, adaptive sub-stepping during close encounters, and a neural network limited to a scalar pairwise force correction. Its novelty is therefore methodological: through controlled ablation, it tests which physical constraints are necessary for dynamically usable long-horizon three-body rollouts.

4 SIMON: Architecture and Methodology

4.1 Overview and Research Objective

The aim of this study is to investigate how closely physically constrained neural network force models can reproduce the long-term stability and energy-conserving behaviour of traditional numerical integrators in three-body simulations, while reducing computational cost without sacrificing qualitative dynamical stability. Rather than attempting to replace established integration schemes, this work seeks to quantify the extent to which neural networks can approximate the physical fidelity of traditional step integrators when computational efficiency is prioritized.

Traditional numerical integrators are expected to achieve superior accuracy, particularly with energy conservation over long timescales, but typically incur higher computational costs. Neural methods, by contrast, offer faster force evaluations but often suffer from less long-term stability. By embedding physical constraints directly into the neural network architecture, this study examines whether such models can narrow the gap in physical accuracy while retaining their computational advantage.

Formally, we address the following research question:

> *How closely can physically constrained neural network force models approximate the long-term stability of traditional numerical integrators in three-body simulations, while reducing computational cost without sacrificing qualitative dynamical stability?*

Building on the gap identified in Section 3, this section defines the specific hybrid architecture used to test that question. The central design choice is to learn only a scalar pairwise force correction while keeping Newtonian force direction and close-encounter timestep refinement explicit. This separates the learned component from the physical constraints, allowing the later ablation study to test which parts of the method are responsible for long-horizon stability.

4.2 Problem Formulation

We study the unrestricted Three-Body problem where masses solely interact through Newtonian gravitational forces. The governing equations (see Subsection 2.2) for the evolution of this system are deterministic, time-reversible, and conserve total energy. As discussed earlier in section Subsection 2.2, a Three-Body system exhibits chaotic motion for generic initial conditions, making it ideal for testing long-term numerical stability in Physically constrained neural networks.

4.3 Baseline Numerical Solver: `Rebound`

Throughout this paper, we will take the ground truth of the Three-Body integration to be the output from `Rebound` [Rein and Liu, 2012], and any output from neural networks will be compared against `Rebound` output. `Rebound` is an N-body integrator that can simulate the motion of particles under the influence of gravity. `Rebound` has several integrators: and the one we will be using in our case will be the `ias15`, Rein and Spiegel [2014] as it is one of the most precise integrators offered by Rebound. It is a high-order adaptive integrator based on Gauss-Radau quadrature, accurate to near machine precision over long integrations [Rein and Spiegel, 2014]. The "15" in `ias15` refers to its 15th-order accuracy, which enables excellent energy conservation without being a symplectic method. Its adaptive timestep control makes it especially reliable for close encounters and long-horizon simulations.

To get an idea of what the model we will train is actually being trained on, observe 2 simulations with trajectories plotted.

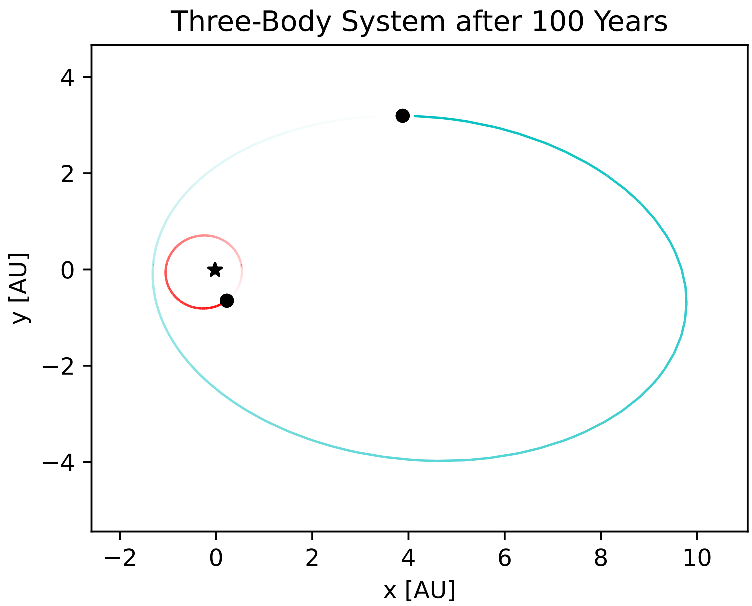

Figure 4: Trajectories calculated by `Rebound`, with body masses $1\,M_{\odot}$, $0.01\,M_{\odot}$, $0.005\,M_{\odot}$, for 100 years.

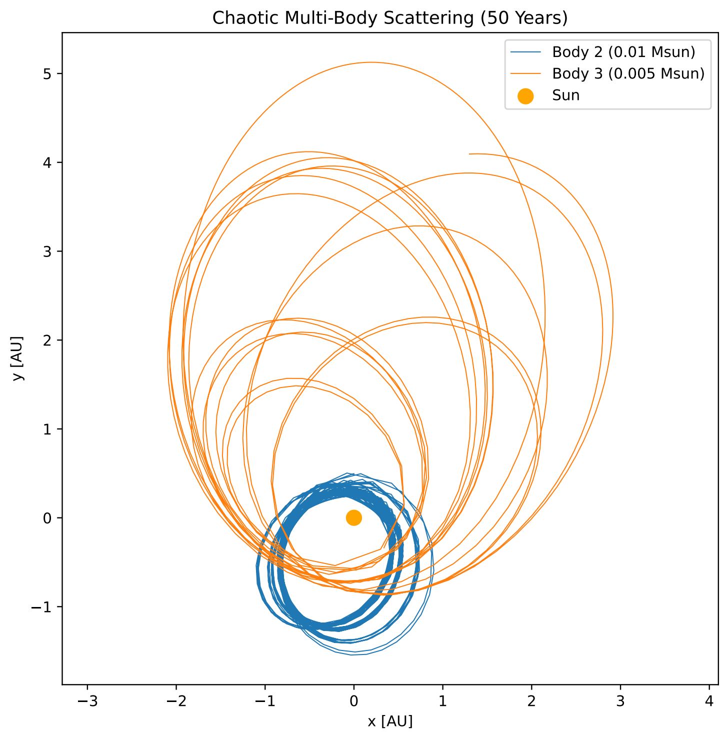

Figure 5: Trajectories calculated by `Rebound`, with body masses $1\,M_\odot$, $0.01\,M_\odot$, $0.005\,M_\odot$, for 100 years, but with different initial positions.

These simulations serve several purposes. First, they establish a physically consistent baseline against which alternative force models can be evaluated. Second, they illustrate the range of orbital behaviors encountered even in relatively simple three-body configurations, including non-circular motion and mutual perturbations. Finally, it provides a concrete visualization of the types of interactions, particularly variations in inter-body separation and acceleration, that any force emulator must accurately capture in order to remain stable over extended integration times.

4.4 Physically Constrained Force Magnification Neural Network

The Pairwise Force Magnitude Network (PFMN) Nguyen et al. [2021], Battaglia et al. [2016], we are training for the purpose of this investigation is not trained to predict forces directly, but instead is trained to predict a scalar pairwise force magnitude, which is essentially the magnitude of the force of interaction between bodies interacting, given in pairs. Let:

- $\mathbf{x}_i(t) \in \mathbb{R}^3$ be the position vector of body i at a time t,
- $\mathbf{v}_i(t) = \dot{\mathbf{x}}_i(t)$ be its velocity vector,

- $m_i > 0$ be its mass,
- G be the universal gravitational constant.

For any pair of bodies, the displacement vector can be given as $r_{ij} = x_j - x_i$. The pairwise distance can be given by $r_{ij} = ||r_{ij}||$. The classical Newtonian equations of motion used for their interactions are; $\sum \mathbf{F}_{ij} = m_i\ddot{\mathbf{x}}$ (second law of motion,) and

$$\mathbf{F}_{ij}^{\text{Newt}} = G\frac{m_i m_j}{|\mathbf{r}_{ij}|^3}\mathbf{r}_{ij}$$

(universal law of gravitation.) Murray and Dermott [1999].

To improve stability during close encounters, a softened gravitational force is also computed using a fixed softening parameter ϵ. This softened force replaces the Newtonian inter-body distance r_{ij} with $\sqrt{r_{ij}^2 + \epsilon^2}$ to ensure forces are controlled during interactions where the inter-body distance r_{ij} is less than ϵ. Dehnen [2001]. This provides a smoother learning target for the model, in the sense that minuscule errors in position do not completely break the simulation. The new, softened force can be represented as $\tilde{r}$. However, when using softening, a slight bias is introduced, which why during model analysis/evaluation we will be using unsoftened forces.

Hence, substituting into the universal law of gravitation, we get the following representation of a pairwise force:

$$\mathbf{F}_{ij}^{\text{soft}} = G\frac{m_i m_j}{\tilde{r}_{ij}^3}\mathbf{r}_{ij}.$$

Now that the mathematics of the method of force formulation has been developed, we will now move onto the model architecture. The learned model is a scalar function with neural network weights θ, $f_\theta : (\tilde{r}_{ij}, m_i, m_j) \mapsto F_{ij} \in \mathbb{R}$, where $\hat{F}_{ij}$ is the predicted pairwise force magnitude. When we implement this, the network will receive a feature vector $z_{ij} = [r_{ij}, m_i, m_j] \in \mathbb{R}^3$, and will output $\hat{F}_{ij} = f_\theta(z_{ij})$. We now construct a unit vector for the direction of the force, given by

$$u_{ij} \equiv \frac{\mathbf{r}_{ij}}{||\mathbf{r}_{ij}||}.$$

Now that we have both the direction of the unit vector and the magnitude, we can multiply $\hat{F}_{ij}$ and u_{ij} to get the magnitude and direction of the force:

$$\mathbf{F}_{ij}^{NN} = F_{ij}\mathbf{u}_{ij}$$

. Now, to get the sum of the forces on body i we will sum over all forces:

$$\mathbf{F}_i^{NN} = \sum_{j \neq i} \mathbf{F}_{ij}^{NN}$$

. Kipf et al. [2018].

Moreover, the architecture is a neural network with 3 hidden layers of 32 neurons each, using SiLU activation functions: Input(3) $\rightarrow$ Linear(32) $\rightarrow$ SiLU $\rightarrow$ Linear(32) $\rightarrow$ SiLU $\rightarrow$ Linear(32) $\rightarrow$ SiLU $\rightarrow$ Linear(1).

4.5 Hybrid Incorporation

Ulibarrena et al. [2024] predicts that using a hybrid combination of an analytical integrator (like `ias15` in our case) to correct inaccurate neural network predictions could "prevent large energy errors." A fixed timestep is used. We incorporate `ias15` into our model in a similar way Ulibarrena et al. [2024] did, replacing inaccurate neural predictions with an analytic one instead. We do this by not letting the NN predict by itself, but rather learn a correction, c, to a softened gravitational force Kipf et al. [2018]. The coefficient of correction, predicted by the NN, determines whether the softened force should be stronger or weaker. We divide force computation into three tiers. Far field, where $r > 500\epsilon = 0.15\,\mathrm{AU}$ means the NN is not invoked and is identical to what `ias15` predicts. Close encounters, where $r < 0.15\,\mathrm{AU}$ means the NN is invoked and predicts a correction factor. To prevent the force from being strengthened or weakened too much, we gate the coefficient, bounding it between 0.2 and 5. If it falls outside of this range, we do not use the NN correction and simply take the softened baseline force. Lastly, for very close encounters, where $r < 5 \times 10^{-4}\,\mathrm{AU}$, the NN is again bypassed to ensure that no bodies are ejected, and non-physical behavior is prevented.

4.6 Training Process

Training data is generated by sampling 50,000 physically valid pairwise configurations from `Rebound`. For each configuration, reference forces are computed analytically using Newtonian gravity. Approximately 60% of the samples are drawn from regions where the inter-body distance $r \in [0.5\epsilon, 50\epsilon]$ ($\epsilon = 3 \times 10^{-4}$), where the correction predicted by the model would deviate meaningfully from 1. The other 40% is sampled from the far field $r \in [50\epsilon, 10]$ where the correction would be closer to one.

Furthermore, the following hyperparameters are used for the model.

- **Optimizer**: AdamW, weight decay $= 1 \times 10^{-5}$
- **Learning rate**: 1×10^{-3}, cosine annealing to 1×10^{-5} Loshchilov and Hutter [2017]

over 5,000 epochs.

- **Loss Function**: MSE on $\log(c)$
- **Batch Size**: Full batch (40,000 training samples)
- **Train/evaluation split**: 80% / 20% (40,000/10,000)
- **Training device**: NVIDIA GeForce RTX 5050 CUDA 13.0
- **Training time**: 9 seconds
- **Best MSE**: 1.1×10^{-5}
- **Training seed**: 42 (see Section 5.3.3 for robustness across seeds)

4.7 Adaptive Sub-Stepping

The integrator uses a fixed macro timestep dt. To resolve close encounters where gravitational acceleration changes faster than dt can capture, an adaptive sub-stepping mechanism is applied Rein and Spiegel [2014]. At each step dt the minimum pairwise distance $r_{\min}$ is computed across all body pairs. If $r_{\min} < 0.05\,\mathrm{AU}$ then dt will be split into smaller sub-steps. The number of sub-steps is determined by the following:

$$n_{\text{sub}} = \min\left(16, \max\left(2, \left\lceil \frac{0.05}{r_{\min}} \right\rceil\right)\right).$$

Each sub-step has a duration $\frac{dt}{n_{\text{sub}}}$. This makes sure that as encounters become closer and closer, the number of sub-steps increases proportionally to account for the increased gravitational force (and hence acceleration) that is associated with closer encounters. The maximum of 16 sub-steps per large dt caps the computational cost to optimize speed.

All physics (positions, velocities, accelerations, forces) is computed in float64 for numerical stability. The NN forward pass uses float32 weights, with the correction factor being converted back to float64 before use.

4.8 Evaluation Strategy

Long-term trajectory divergence in chaotic systems like the Three-Body problem is inevitable, as established in section Subsection 2.2. With this knowledge in mind, we will evaluate with the following diagnostics on the hybrid model:

- **Trajectory Overlay:** Trajectories generated by neural models are superimposed against the reference numerical solution to gain qualitative insight into the accuracy of the model. This will be done with various different initial conditions to explore how the model will handle different situations, to get a better insight into its versatility. Moreover, we will also plot a time series overlay to analyze if there is any phase drift.

- **Trajectory Divergence:** To gain insight into the sensitivity of each system with respect to the initial conditions, we numerically estimate the trajectory divergence by evolving predicted trajectories in parallel, then calculating the RMS position error between the two. Agreement between neural and reference values indicates preservation of the system's chaotic structure.
- **Speed-Accuracy Tradeoff:** The distance between the neural network predicted trajectory and the `Rebound` predicted trajectory, plotted over throughput (how fast trajectory points were saved.)

4.9 Scope of the Method

This method is designed to see how closely a physically constrained Neural Network can shadow a traditional iterative integrator in terms of close counters, computational efficiency, and trajectory divergence. By comparing these two, we seek insight into the development of neural methods of Three-Body emulation, and learn to what extent these methods are able to speed-up simulations without compromising significant accuracy and stability.

5 Results

In this section we present the experimental results for `SIMON` across four themes. Section 5.1 isolates the individual contribution of each architectural component through a controlled four-run ablation. Section 5.2 evaluates generalisation across seven initial conditions. Section 5.3 examines physical consistency and reproducibility. Section 5.4 characterises the speed-accuracy trade-off as a separate computational-performance result.

Throughout this section, λ denotes the fitted divergence-rate proxy (slope of $\log \delta(t)$ in the 10–50 yr window), and "bounded" indicates that all three bodies remain within 10 AU of the system barycentre at $T = 100\,\text{yr}$. The reference integrator is `ias15` [Rein and Spiegel, 2014]; all initial conditions use masses [1.0, 0.01, 0.005] $M_\odot$ unless otherwise stated. At the operational timestep $dt = 0.04\,\text{yr}$, $\lambda = 0.1655\,\text{yr}^{-1}$ and final RMS $= 1.54\,\text{AU}$ for the default initial condition (IC1).

5.1 Results I — Stability Decomposition

5.1.1 The Four-Run Ablation Design

To isolate the role of each architectural component, we conduct a controlled four-run ablation in which exactly one component is removed or altered per run while the initial condition IC1, timestep grid, simulation horizon, and evaluation protocol remain fixed. Runs A, C, and D use the same trained SIMON weights; Run B uses a separately trained Vector NN baseline because the architecture itself is changed to learn force direction. The four runs are summarised in Table 1.

Table 1: Four-run ablation framework. Each run removes or replaces exactly one component. Run D is the full SIMON system and serves as the reference configuration. All runs use IC1 and $T = 100$ yr under the same evaluation protocol. Run A reports the adaptive-off failure observed in the timestep-grid sweep, where $dt = 0.01$ yr produces ejection. Runs A, C, and D use the same SIMON weights; Run B uses a separately trained Vector NN baseline because the force-direction architecture is changed.

Run	Adaptive stepping	Force direction	NN correction	Outcome
A	**OFF**	Analytic	ON	Ejection (161.7 AU)
B	ON	**Learned (Vector NN)**	N/A	Ejection (83.0 AU)
C	ON	Analytic	**OFF**	Bounded (1.67 AU)
D (SIMON)	ON	Analytic	ON	Bounded (1.54 AU)

The logic of the design is as follows. Runs A and B establish stability requirements by showing that removing a component causes ejection. Run C establishes the NN's role by showing that its removal leaves the system stable. Run D is the full system, providing the accuracy ceiling against which Run C is compared. Because Run B is evaluated on the default initial condition, its trajectory outcome is interpreted together with the gate-sensitivity analysis rather than as a standalone statistical generalisation.

5.1.2 Run A: Adaptive Sub-Stepping Is the Primary Stability Mechanism

Without adaptive sub-stepping, `SIMON` uses a fixed leapfrog step for every integration step regardless of body separation. At $dt = 0.01$ yr, a close encounter occurs whose duration is shorter than the fixed timestep can resolve. The lightest body receives an incorrectly large gravitational impulse in a single step and is ejected to 161.7 AU at $T = 100$ yr. With adaptive sub-stepping active, the same close encounter is resolved by splitting the step into 2 to 16 sub-steps when $r_{\min} < 0.05$ AU, and ejection is eliminated across all tested timesteps for IC1.

Figure 6 shows the complete frontier sweep across five timestep values: $dt = 0.005$, 0.01, 0.02, 0.04, and 0.08 yr. The sweep is run for both conditions. The worst-case final RMS across the frontier reduces from 161.7 AU before adaptive sub-stepping ($dt = 0.01$ yr) to 1.54 AU after adaptive sub-stepping ($dt = 0.04$ yr), a 105× improvement.

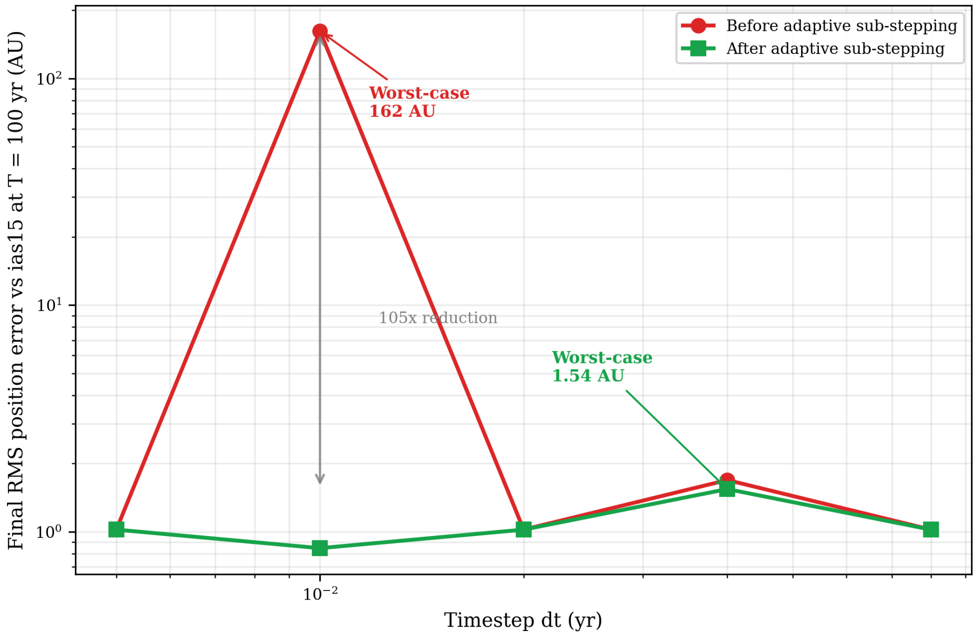

Figure 6: Final RMS position error vs. timestep dt for `SIMON` with (green squares) and without (red circles) adaptive sub-stepping. Without adaptive stepping, $dt = 0.01$ yr causes body ejection to 161.7 AU. With adaptive stepping, all tested timesteps in this IC1 frontier sweep produce bounded dynamics. Worst-case improvement across the frontier: 105× (161.7 → 1.54 AU).

At the operational timestep $dt = 0.04$ yr—the main timestep used for the operational comparison—adaptive sub-stepping costs +28% per-step overhead (19.21 → 24.57 μs), reducing the speedup from 1.53× to 1.12× relative to ias15. Both configurations remain faster than `ias15`. The reported NN invocation rate is 0.45–1.08% of pair evaluations involving close encounters with $r < 0.15$ AU. Adaptive sub-stepping uses the stricter condition $r_{\min} < 0.05$ AU, so it is triggered only on a subset of those close-encounter cases, leaving the majority of steps at the base leapfrog cost.

5.1.3 Run B: Analytic Force Direction as a Structural Physical Constraint

In Run B, the analytic force direction $\mathbf{u}_{ij} = \mathbf{r}_{ij}/\|\mathbf{r}_{ij}\|$ is replaced by a learned direction from a Vector NN. The Vector NN uses the same hidden-layer structure as `SIMON` but receives strictly more information, including the geometric direction components $[r_x/r, r_y/r, r_z/r]$, and predicts a unit force-direction vector. Post-training verification confirms sub-degree directional accuracy, with maximum directional error approximately 0.04° across the tested distance range. For the tested initial condition, however, the learned-direction architecture produces unbounded trajectories, reaching final RMS error 83.0 AU at $T = 100$ yr, while `SIMON` remains bounded at 1.54 AU. In plain terms, this run tests whether gravity's direction should be learned approximately or enforced exactly from geometry.

To test whether this unbounded outcome was caused by a single anomalous direction prediction or by accumulated small errors, the Vector NN direction gate was tightened and the simulation was re-run.

Table 2: Gate-sensitivity analysis for the learned-direction Vector NN baseline. Tightening the gate from the standard 30° threshold to a sub-degree threshold does not change the outcome, indicating that the failure is not caused by a single large outlier prediction.

Gate setting	Interpretation	Outcome
$\cos(\theta) < 0.866$	Standard 30° gate; triggers only on large direction deviations.	Ejection unchanged; final RMS 83.0 AU.
$\cos(\theta) < 0.9999985$	Much tighter than the standard gate; close to the Vector NN's measured sub-degree accuracy.	Identical ejection timing; final RMS 83.0 AU.

The gate-sensitivity result shows that the mechanism is not a single catastrophic misprediction, but the accumulation of small non-radial impulses over many steps. When the gate is tightened to $\cos(\theta) < 0.9999985$, corresponding to an angular threshold of approximately 0.1°, the ejection timing and final RMS remain unchanged. This threshold is far tighter than the original 30° gate and close to the Vector NN's measured sub-degree directional accuracy.

Thus, the failure is not an outlier that a gate can catch. Instead, many individually acceptable direction errors accumulate during close encounters. A simple scale estimate illustrates the issue: at $dt = 0.04\,\mathrm{yr}$ over 100 yr, there are approximately 2,500 integration steps, and 0.04° per close-encounter step can accumulate to roughly 100° of non-radial impulse. Because Newtonian gravity is a central force directed exactly along the line connecting two bodies, any learned perpendicular component has no physical basis and can compound into non-physical energy injection in a chaotic system.

Figures 7 and 8 illustrate the physical consequence of the gate-sensitivity finding. During the bounded phase, `SIMON` and the Vector NN have nearly identical fitted divergence rates (0.1655 vs. $0.1656\,\mathrm{yr}^{-1}$), but the learned-direction model later becomes unbounded for this tested initial condition, reaching final RMS error 83.0 AU. In Figure 8, the long red dotted segments are escape paths of the Vector NN trajectory, not stable orbital loops.

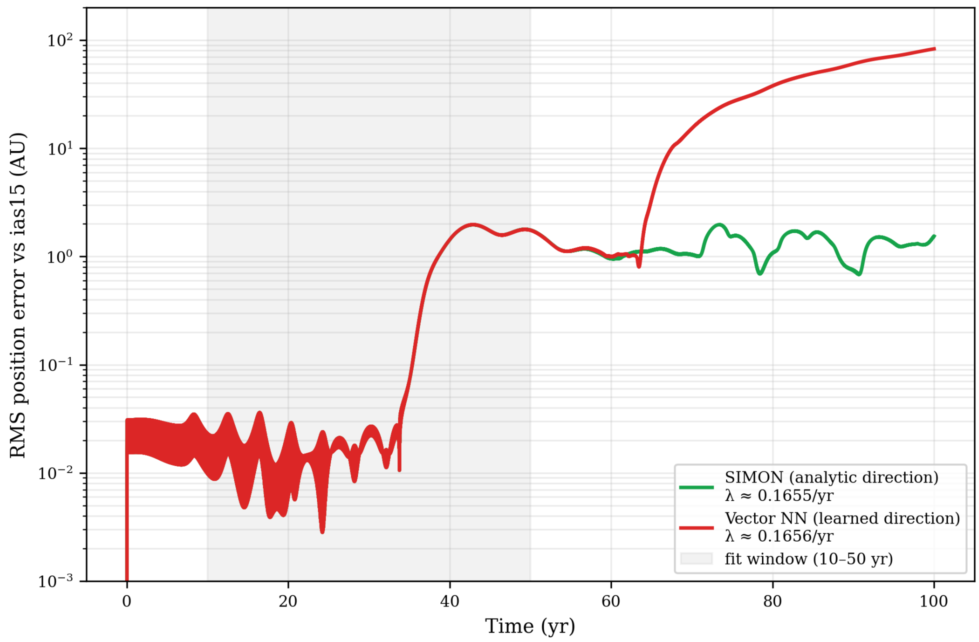

Figure 7: RMS position error vs. time for `SIMON` (green, analytic direction, $\lambda \approx 0.1655\,\mathrm{yr}^{-1}$, bounded) and Vector NN (red, learned direction, $\lambda \approx 0.1656\,\mathrm{yr}^{-1}$ in the fit window but later unbounded). Fit window (grey shading): 10–50 yr. The nearly identical fitted divergence rates show that both models reproduce similar bounded-phase chaotic divergence; the later unbounded Vector NN trajectory illustrates the accumulated non-radial impulse mechanism identified by the gate-sensitivity analysis.

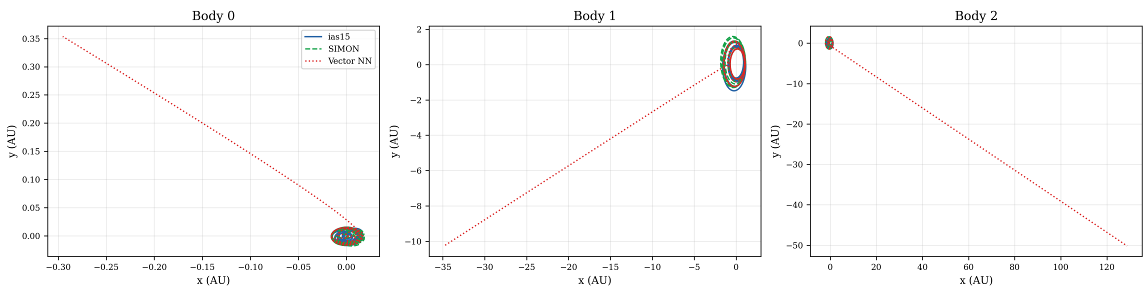

Figure 8: XY trajectory overlay for all three bodies over 100 years: `ias15` (blue solid), `SIMON` (green dashed), Vector NN (red dotted). Each panel uses its own axis limits. For the tested initial condition, `ias15` and `SIMON` remain bounded, while the learned-direction Vector NN develops long red dotted escape paths, most clearly for Bodies 1 and 2. This figure illustrates the physical consequence of the gate-sensitivity mechanism rather than serving as a standalone statistical generalisation.

This result supports the conclusion that analytic force direction is not merely a convenience but a structural stability constraint. For the tested initial condition, the learned-direction baseline becomes unbounded despite sub-degree angular accuracy, and the gate-sensitivity analysis shows that the outcome is insensitive to the specific gate threshold chosen. Analytic direction therefore removes, by construction, a non-physical perpendicular-force error mode that a learned direction model can still introduce during close encounters.

5.1.4 Run C: The Neural Network Correction Is an Accuracy Mechanism

In Run C, the NN correction is disabled entirely: the correction factor c is fixed at 1.0 for all pair evaluations, reducing the force to pure softened Newtonian gravity. The analytic force direction and adaptive sub-stepping remain active. The result is a bounded simulation at 1.67 AU—not an ejection.

Figure 9 shows that Run C (No-NN baseline, red dashed) and Run D (full `SIMON`, green solid) are visually indistinguishable throughout the 100-year horizon. Both achieve $\lambda = 0.1655\,\mathrm{yr}^{-1}$ to four decimal places. The NN correction contributes 8.5% improvement in final RMS accuracy ($1.67 \to 1.54$ AU) at the 0.45% of pair evaluations involving close encounters below 0.15 AU, where it is invoked.

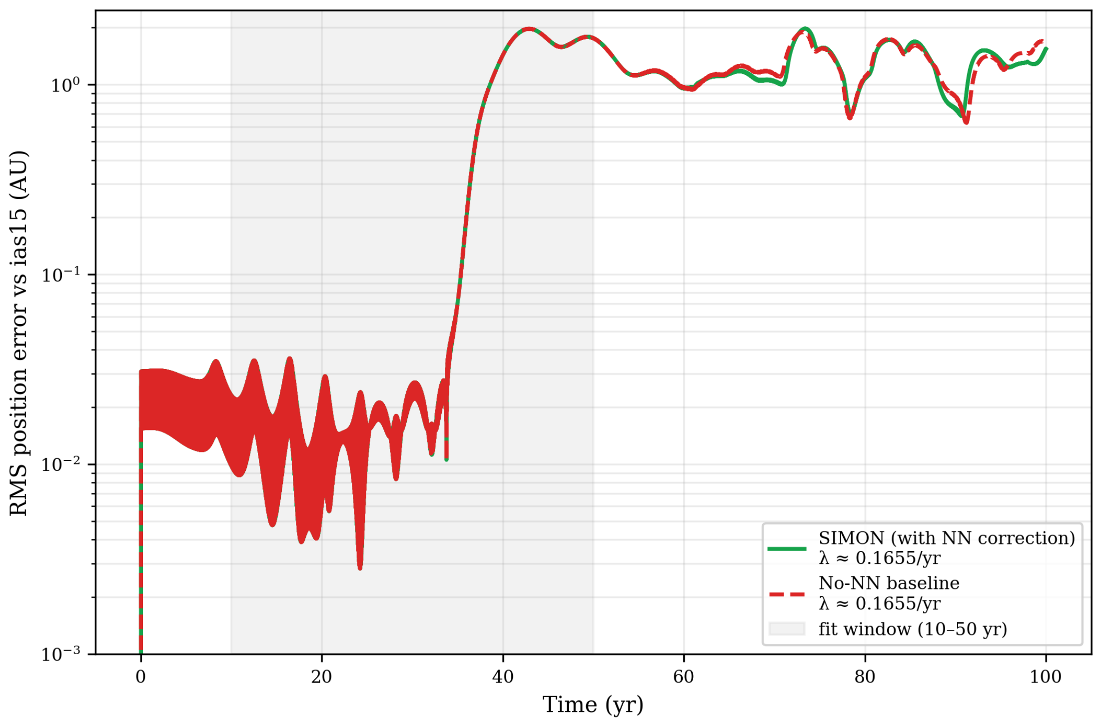

Figure 9: RMS position error vs. time for `SIMON`/Run D (green solid, $\lambda \approx 0.1655\,\mathrm{yr}^{-1}$, 1.54 AU final RMS) and the No-NN baseline/Run C (red dashed, $\lambda \approx 0.1655\,\mathrm{yr}^{-1}$, 1.67 AU final RMS). Both curves are bounded and nearly identical. The 8.5% accuracy improvement from the NN correction is the only measurable difference.

This result clarifies the role of the NN correction. Run A shows that close-encounter stability is provided by adaptive sub-stepping, since removing that mechanism causes ejection even when the NN correction remains active. Run C shows that removing the

NN correction while retaining adaptive sub-stepping leaves the system bounded, but increases final RMS error from 1.54 AU to 1.67 AU. Therefore, the NN correction is best interpreted as a targeted accuracy refinement at close-encounter steps, rather than the primary stability mechanism.

5.1.5 Synthesis: The Two-Constraint Principle

The four-run ablation establishes a clear decomposition of `SIMON`'s performance:

- **Adaptive sub-stepping** eliminates the leapfrog ejection failure at $dt = 0.01$ yr for IC1 (Run A: 161.7 AU without sub-stepping versus 0.847 AU with sub-stepping).
- **Analytic force direction** is a structural physical constraint (Run B: for the tested initial condition, replacing analytic direction with a learned direction produces unbounded trajectories with final RMS 83.0 AU).
- **NN correction** is an accuracy mechanism (Run C: removing it leaves stability intact but costs 8.5% accuracy).
- **Full** `SIMON` (Run D) achieves the best accuracy by combining all three components.

Figure 10 shows the XY trajectory overlay and coordinate time series for the full `SIMON` system (Run D), confirming bounded dynamics for all three bodies over 100 years.

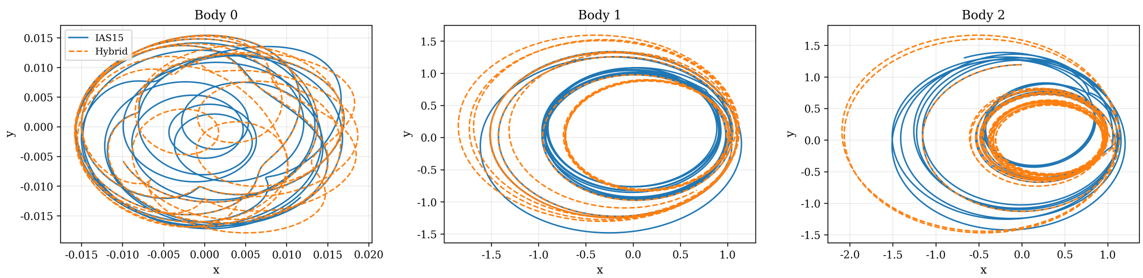

Figure 10: XY trajectory overlays for all three bodies over 100 years comparing `SIMON` (dashed) to `ias15` (solid). All three bodies remain gravitationally bounded throughout the full 100-year horizon.

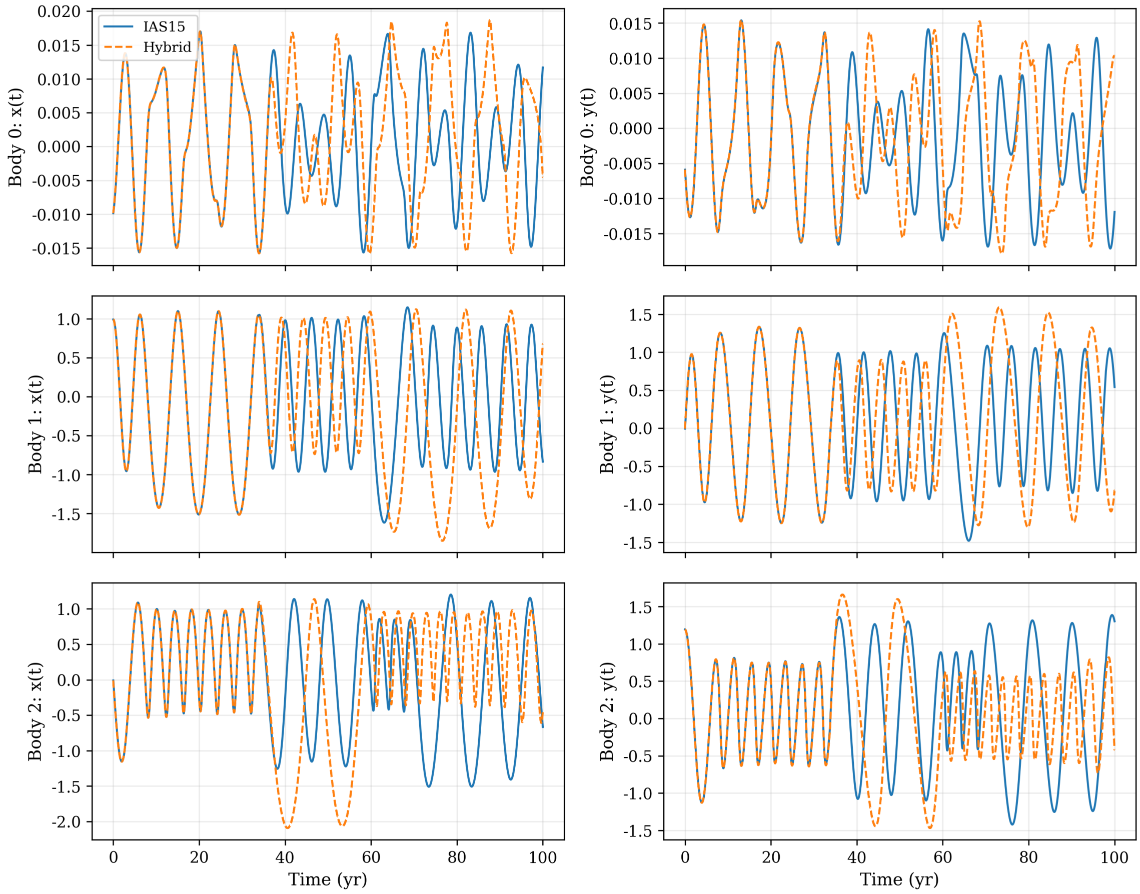

Figure 11: Coordinate time series $x(t)$, $y(t)$ for all three bodies over 100 years. Early-time agreement is strong; mismatch accumulates as accumulated phase drift after 25–30 years, consistent with the chaotic divergence of the Three-Body Problem. All coordinates remain oscillatory and bounded within ranges consistent with `ias15`.

The primary error mode is phase drift—a gradual timing mismatch that accumulates while preserving the qualitative orbital geometry and boundedness of all three bodies. This is consistent with the chaotic sensitivity of the Three-Body Problem [Ott, 2002] and does not indicate structural failure of the integrator.

5.2 Results II — Generalisation Across Initial Conditions

5.2.1 Seven Initial Conditions Spanning the Dynamical Spectrum

To test whether `SIMON`'s performance extends beyond the single default configuration (IC1) used in the ablation study, we evaluate across seven distinct three-body initial conditions designed to span qualitatively different dynamical regimes. Table 3 summarises the configuration and simulation outcomes for all seven ICs. All runs use masses $[1.0, 0.01, 0.005]\, M_\odot$ except IC2 (near-equal mass), $T = 100\,\mathrm{yr}$, and $dt = 0.04\,\mathrm{yr}$.

Table 3: Multi-IC evaluation results ($T = 100\,\text{yr}$, $dt = 0.04\,\text{yr}$). Mean and standard deviation of λ and final RMS are computed over the four bounded ICs only. IC2, IC5, and IC7 are physically unstable (confirmed independently by `ias15`; see Section 5.2.3).

IC	Regime	λ (yr^{-1})	RMS (AU)	Bounded	Speedup
IC1: Default	Moderate scattering	0.1655	1.54	Yes	1.11×
IC2: Near-equal	Strongly interacting	0.0430	116.9	No	1.69×
IC3: Tight pair	High encounter rate	0.1207	3.45	Yes	1.27×
IC4: Hierarchical	Weakly coupled outer	0.0990	1.60	Yes	1.12×
IC5: High ecc. (v1)	High eccentricity	0.0648	20.0	No	1.17×
IC6: Near-circular	Low chaos	−0.019	0.008	Yes	1.07×
IC7: High ecc. (v2)	High eccentricity	0.0341	11.4	No	1.07×
Mean (bounded ICs only)		0.092	1.65	–	–
Std (bounded ICs only)		0.068	1.22	–	–
Bounded count		–	–	4/7	–

5.2.2 Bounded Configurations: Four ICs

All four stable configurations maintain bounded dynamics for the full 100-year horizon. The configurations span the complete stable dynamical spectrum of the three-body problem:

- **IC6 (near-circular)**: $\lambda \approx 0$, final RMS $= 0.008\,\text{AU}$. The NN is never invoked (NN_frac $= 0.000$); `SIMON` computes exact Newtonian forces for every step. The residual 0.008 AU is the pure leapfrog–Gauss-Radau integrator difference.
- **IC4 (hierarchical)**: $\lambda = 0.099\,\text{yr}^{-1}$, final RMS $= 1.60\,\text{AU}$. The outer body at 5 AU is weakly coupled; the NN is also never invoked.
- **IC1 (default)**: $\lambda = 0.1655\,\text{yr}^{-1}$, final RMS $= 1.54\,\text{AU}$. The default IC1 baseline.
- **IC3 (tight binary)**: $\lambda = 0.121\,\text{yr}^{-1}$, final RMS $= 3.45\,\text{AU}$. The inner body at 0.5 AU generates continuous close encounters; despite this, `SIMON` remains bounded throughout.

Figure 12 summarises the divergence rates, final RMS values, and speedups across all seven ICs.

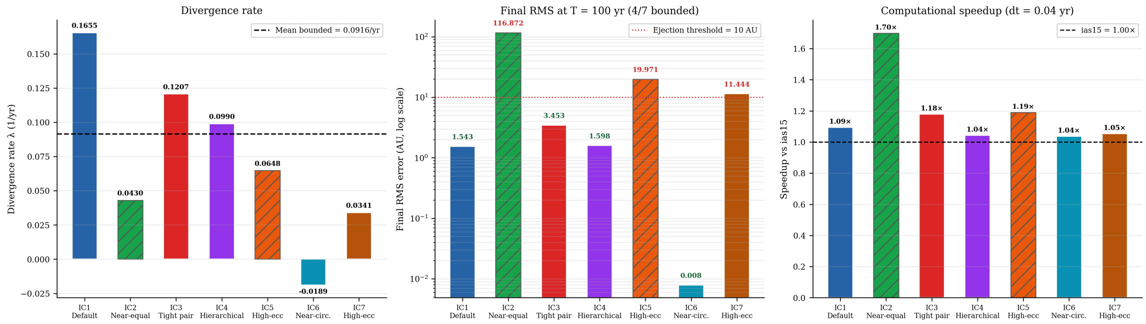

Figure 12: Multi-IC evaluation summary. Left: fitted divergence rate λ for all 7 ICs; dashed line = mean of bounded ICs. Centre: final RMS at T = 100 yr (log scale); dotted line = 10 AU ejection threshold. Right: speedup vs. `ias15`. Hatched bars (IC2, IC5, IC7) denote physically unstable configurations confirmed by `ias15`.

Figure 13 shows the divergence curves for all seven ICs simultaneously.

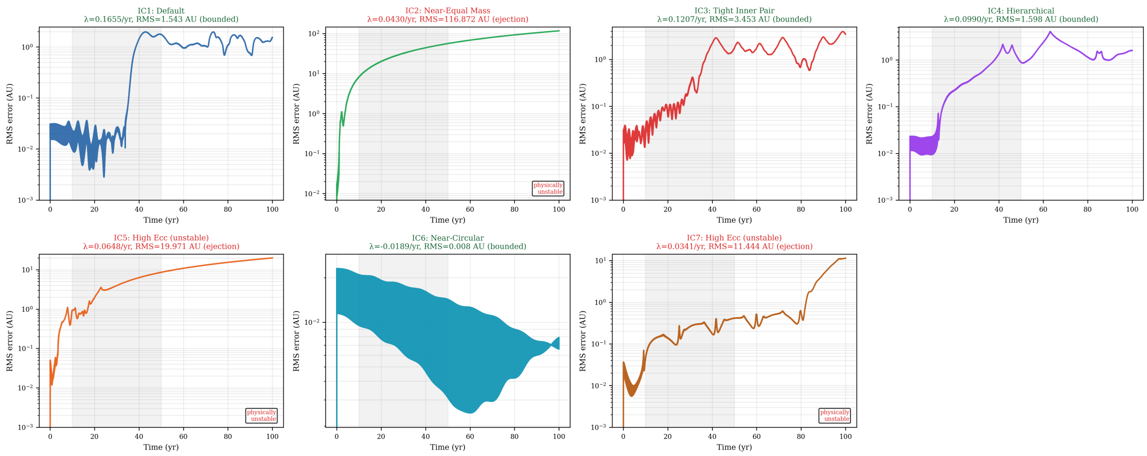

Figure 13: RMS position error vs. time for all 7 initial conditions (2×4 panel). Grey shading: 10–50 yr fit window. Green/teal labels: bounded configurations. Red labels: ejection cases. The four bounded ICs maintain smooth exponential divergence characteristic of chaotic but stable three-body dynamics.

5.2.3 Physically Unstable Configurations

Three configurations produce ejections (IC2, IC5, IC7). In each case, the reference integrator `ias15` independently also ejects a body, confirming these are physical instabilities rather than `SIMON` failures:

- **IC2 (near-equal mass)**: `SIMON` ejects to 116.9 AU; `ias15` ejects to 58.9 AU. The 2× post-ejection discrepancy is consistent with chaotic amplification of integrator differences after the system becomes dynamically unbound.
- **IC5 (high eccentricity, v = 2.2, r = 0.3 AU)**: Both integrators eject; `ias15` ejects Body 2 at t = 34.8 yr.

- **IC7 (high eccentricity redesign, $v = 1.6$, $r = 0.6$ AU)**: Both integrators eject; `ias15` ejects Body 2 at $t = 88.4$ yr.

`SIMON` does not suppress these physical instabilities. This is expected and physically consistent: the model agrees with `ias15` on the occurrence of dynamical instability rather than artificially stabilising configurations that are physically unbound. The post-ejection distances are not expected to match exactly because chaotic divergence amplifies integrator differences after unbinding. High-eccentricity configurations with these mass ratios proved intrinsically unstable in both cases tested: the eccentric inner body transfers orbital energy to the outer body through repeated periapsis passages, eventually unbinding it regardless of integrator.

5.3 Results III — Physical Consistency and Reproducibility

5.3.1 Energy Conservation: Leapfrog Attribution

We track fractional energy drift $\Delta E(t)/E_0 = (E(t) - E_0)/|E_0|$ throughout each simulation, where $E(t) = \mathrm{KE}(t) + \mathrm{PE}(t)$ is the total mechanical energy. `ias15` conserves energy to float64 machine precision ($\approx 10^{-14}\%$). `SIMON` at the operational timestep ($dt = 0.04$ yr) shows:

- IC1: 1.39% maximum fractional drift (NN_frac = 0.0045)
- IC3: 0.67% maximum fractional drift (NN_frac = 0.000)
- IC4: 13.77% maximum fractional drift (NN_frac = 0.000)

IC3 and IC4 both have NN_frac = 0.000: the NN correction was never applied during their simulations. Yet both show comparable energy drift to IC1. The NN correction cannot cause energy drift it never contributed to; the drift must arise from the leapfrog integrator. This is the attribution proof.

IC4's higher fractional drift (13.77%) reflects its small binding energy $|E_0| = 0.006$ ($\mathrm{AU}^2/\mathrm{yr}^2$)—the smallest of the three ICs tested. Because fractional drift is normalised by $|E_0|$, weakly bound hierarchical configurations can show larger relative energy drift at the same timestep. This should not be interpreted as NN-driven energy injection: the NN was never invoked for IC4 (NN_frac = 0.000). Importantly, the drift is oscillatory throughout—not monotonically increasing—consistent with symplectic shadow-Hamiltonian behaviour, and all bodies remain bounded (IC4 final RMS = 1.60 AU).

Figure 14 shows the energy drift time series for all three ICs.

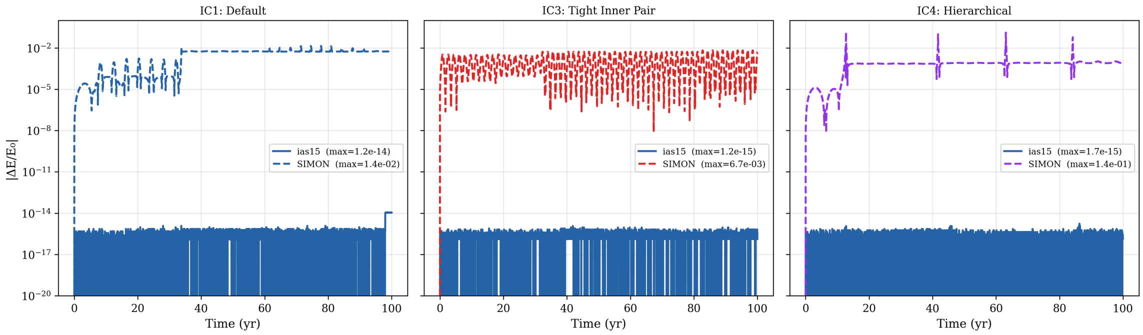

Figure 14: Fractional energy drift $|\Delta E/E_0|$ over 100 years for `ias15` (blue) and `SIMON` (coloured dashed) at $dt = 0.04$ yr across three initial conditions. `ias15` conserves energy to $\approx 10^{-14}\%$ in all cases. `SIMON`'s drift is oscillatory (not monotonic) and attributable to the leapfrog integrator: IC3 and IC4 have NN_frac = 0.000 yet show comparable drift to IC1.

5.3.2 Timestep Scaling: Quantitative Confirmation of dt^2 Prediction

Second-order leapfrog theory predicts that energy drift should scale as dt^2: halving the timestep should reduce drift by a factor of four. We test this by running `SIMON` on IC1, IC3, and IC4 at two timestep values ($dt = 0.01$ yr and $dt = 0.04$ yr). The fitted log-log slopes are:

- IC1: slope = 1.32 (NN_frac = 0.0045)
- IC3: slope = 2.43 (NN_frac = 0.000)
- IC4: slope = 1.86 (NN_frac = 0.000)
- Average: 1.87 (theoretical prediction: 2.0)

The average slope of 1.87 is consistent with dt^2 scaling, given that only two data points are available per IC. For IC4 specifically, the drift reduces from 13.77% at $dt = 0.04$ yr to 1.05% at $dt = 0.01$ yr—a factor of 13.2×, close to the dt^2 prediction of 16×. IC3 and IC4 follow this scaling despite NN_frac = 0.000 at both timesteps, confirming that their energy drift is controlled by the leapfrog timestep rather than by the neural correction.

Figure 15 shows the log-log scaling plot.

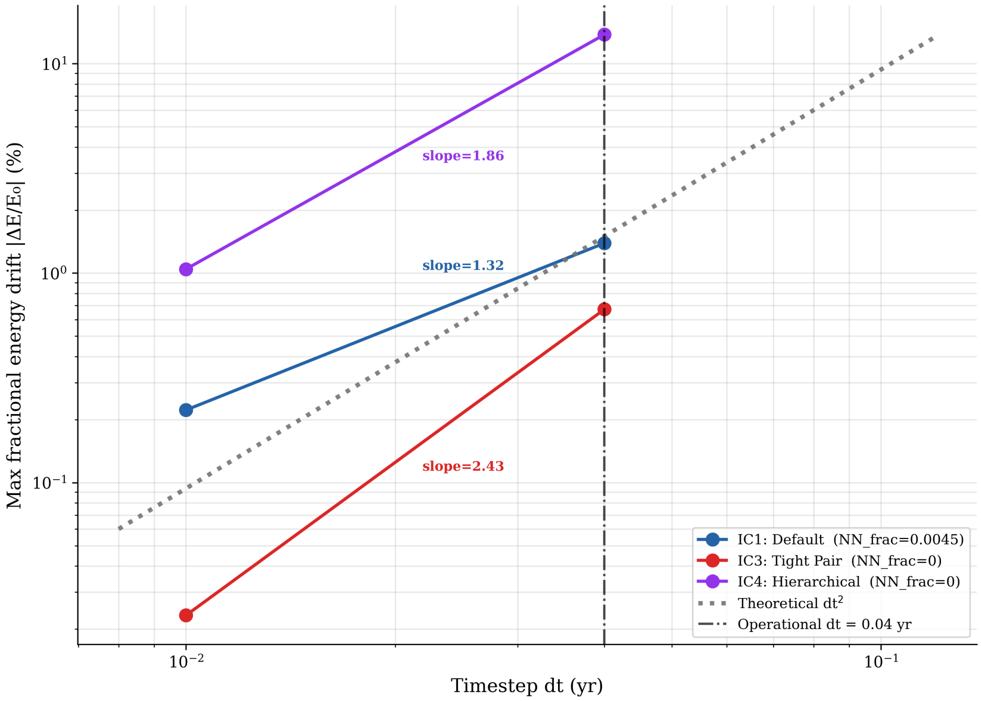

Figure 15: Log-log plot of maximum fractional energy drift vs. timestep dt for IC1 (blue), IC3 (red), and IC4 (purple). Only $dt = 0.01$ and $dt = 0.04\,\text{yr}$ are valid data points (see Section 5.3.2). Dotted lines show fitted slopes (1.32, 2.43, 1.86). Grey dotted reference: theoretical dt^2 slope of 2.0. Vertical dashed line: operational $dt = 0.04\,\text{yr}$.

5.3.3 Training Seed Robustness

To verify that results from the seed-42 model used throughout this paper are not specific to a particular random initialisation, training was repeated five times with seeds $\{42, 123, 456, 789, 1000\}$. Table 4 summarises the training and simulation outcomes.

Table 4: Training seed robustness. All seeds use identical hyperparameters and training data. Simulation results are from the full `SIMON` run on IC1, $T = 100\,\text{yr}$, $dt = 0.04\,\text{yr}$.

Seed	**Val. MSE**	$\boldsymbol{\lambda}$ (yr^{-1})	**Bounded**
42 (primary model)	1.11×10^{-5}	0.1655	Yes
123	1.72×10^{-5}	0.1655	Yes
456 (worst)	1.75×10^{-5}	0.1655	Yes
789	1.68×10^{-5}	0.1655	Yes
1000	1.43×10^{-5}	0.1655	Yes
Mean	1.54×10^{-5}	0.1655	5/5
CV	15.8%	0%	—

The divergence rate $\lambda = 0.1655\,\text{yr}^{-1}$ is identical to four decimal places across all five seeds. All five independently trained models produce bounded dynamics with identical fitted divergence rate, confirming that downstream stability is insensitive to random initialisation.

The 15.8% coefficient of variation in validation MSE does not imply dynamical sensitivity to seed choice: the variation is concentrated at sub-softening distances ($r < 5 \times 10^{-4}$ AU) where the safety gate bypasses NN predictions entirely. In the operationally relevant range ($r \geq 10^{-3}$ AU), all seeds produce correction factor errors below 0.3%—functionally identical—and the simulation metrics are invariant.

Figure 16 shows the training convergence and per-seed validation MSE.

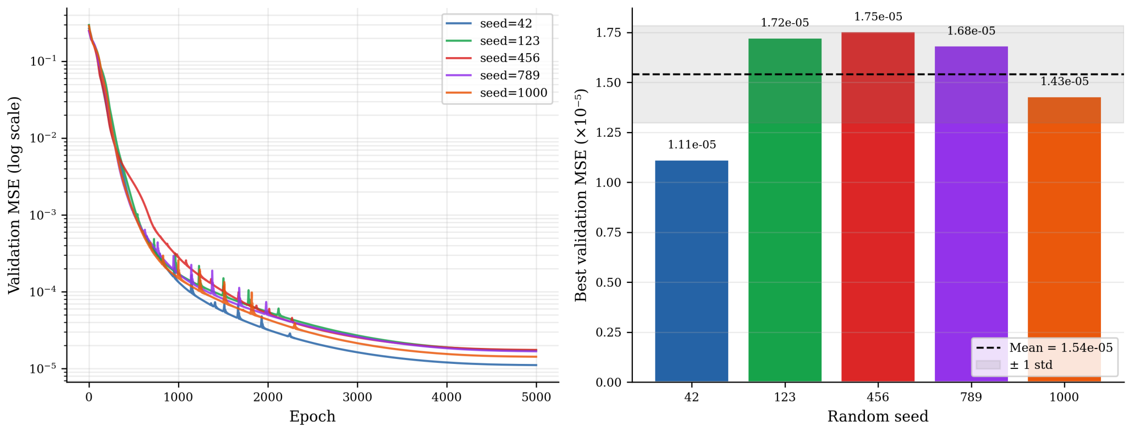

Figure 16: Training seed robustness. Left: validation MSE curves over 5,000 epochs for all five seeds (log scale). All seeds converge smoothly. Right: best validation MSE per seed as a bar chart; dashed line = mean. Despite a 15.8% coefficient of variation in MSE, downstream boundedness and fitted divergence rate λ are invariant across all seeds; the best–worst seed comparison also shows no material change in final RMS.

5.4 Results IV — Computational Performance

5.4.1 Speed-Accuracy Frontier

In this measurement, we observe the final position deviation of the bodies predicted by `SIMON`, in comparison with `ias15`, over a 100 year time horizon. The x-axis is the throughput, which is how often the current state of the system is sampled, `Rebound`, and the y-axis is RMS position error, on a logarithmic scale. The total number of samples is fixed across all simulations. The timestep dt is the timestep size the hybrid model uses internally. Points toward the lower-right of the graph are better (higher throughput, lower error). Throughput is computed as $\frac{n_{\text{samples}}}{t_{\text{total}}}$.

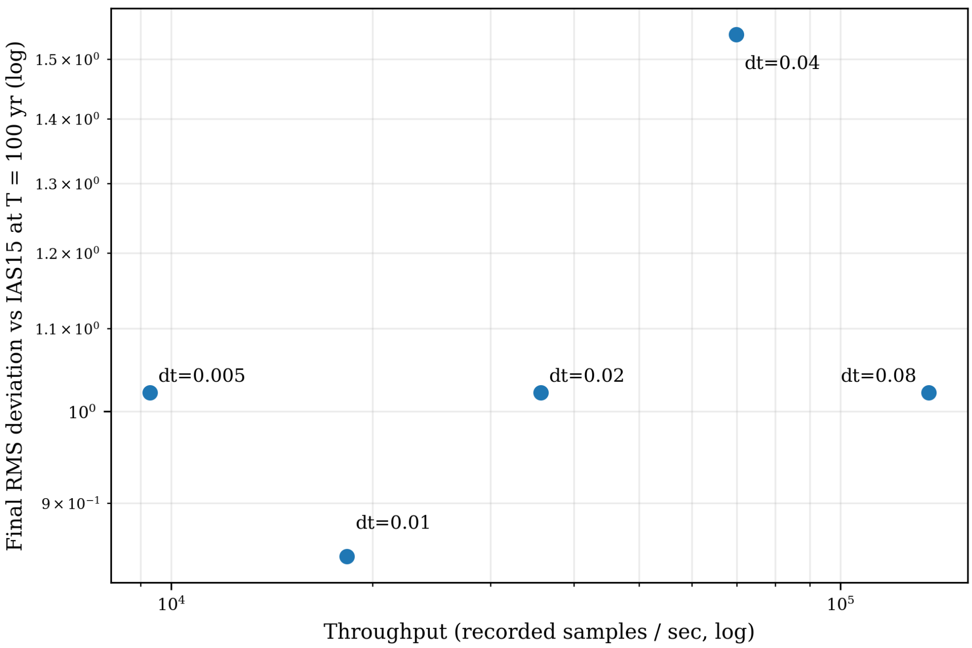

Figure 17: Speed–accuracy trade-off across timesteps dt for `SIMON` with adaptive sub-stepping. Points towards the lower-right are better (higher throughput, lower error). `SIMON` outperforms `ias15` at $dt = 0.04$ ($1.12\times$ speedup) and $dt = 0.08$ ($2.18\times$ speedup). The lowest final RMS error is 0.847 AU at $dt = 0.01$.

The `ias15` baseline completes in 0.080 seconds. `SIMON` outperforms `ias15` at $dt = 0.04$ ($1.12\times$ speedup) and $dt = 0.08$ ($2.18\times$ speedup). The lowest final RMS error is 0.847 at $dt = 0.01$. This is the optimal point for accuracy, but it is approximately $3.4\times$ slower than `ias15`. The errors at $dt = 0.005$, 0.02, and 0.08 are all approximately 1.02, reflecting the chaotic saturation scale. After 100 years, any two different integrations of the same system will diverge to roughly one orbital radius, regardless of timestep. $dt = 0.04$ shows a slightly higher error of 1.54, which is a chaotic fluctuation (a different phase of the orbit was reached at $T = 100$).

Table 5 provides the full quantitative results for the frontier sweep.

Table 5: Speed-accuracy frontier results for `SIMON` with adaptive sub-stepping, $T = 100$ yr. `ias15` baseline: 0.080 s total, $\lambda = 0.1655\,\text{yr}^{-1}$ at $dt = 0.04$ yr. Bold row = operational timestep.

dt (yr)	Final RMS (AU)	Throughput (samp/s)	Time/step (μs)	Total time (s)	Steps	Speedup
0.005	1.022	9,285	26.9	0.538	20,000	$0.15\times$
0.010	0.847	18,307	27.3	0.273	10,000	$0.29\times$
0.020	1.022	35,672	28.0	0.140	5,000	$0.57\times$
0.040	**1.543**	**69,880**	**28.6**	**0.072**	**2,500**	$\mathbf{1.12\times}$
0.080	1.022	135,629	29.5	0.037	1,250	$2.18\times$

The per-step time ranges from 26.9 μs at $dt = 0.005$ to 29.5 μs at $dt = 0.08$: a range of only $2.6\,\mu$s across a $16\times$ change in timestep. This nearly flat profile indicates that the cost of an individual SIMON step is approximately constant and is dominated by force evaluation, pairwise distance checks, and occasional adaptive sub-stepping. Therefore, the observed runtime improvement at larger dt comes primarily from taking fewer macro steps, not from making each individual step substantially cheaper. For a fixed $T = 100\,\text{yr}$ horizon, the number of macro steps scales approximately as T/dt, so total runtime scales roughly as (T/dt) times the near-constant per-step cost.

6 Discussion and Conclusion

6.1 The Two-Constraint Principle and Its Implications

The central finding of this work is that long-horizon orbital stability in a hybrid neural three-body integrator requires two complementary physical constraints acting together:

1. **Adaptive sub-stepping** resolves close encounters that the fixed leapfrog timestep cannot capture. In the timestep-frontier sweep, without adaptive sub-stepping, a single under-resolved encounter injects a large spurious velocity impulse that ejects a body at $dt = 0.01$ yr (Run A: 161.7 AU). This is the primary stability mechanism for this close-encounter failure mode.

2. **Analytic force direction** enforces the exact radial symmetry of Newtonian gravity at every step. In the learned-direction baseline tested in Run B, small accumulated non-radial impulses shift the orbital angular momentum until the trajectory becomes unbounded (final RMS 83.0 AU).

The neural network correction (Run C) does not contribute to stability: removing it while retaining adaptive sub-stepping and analytic force direction produces bounded dynamics at 1.67 AU. The NN's role is a targeted accuracy refinement—8.5% improvement in final RMS position error—at the rare close pair evaluations where the NN correction is invoked. The ablation evidence shows that close-encounter stability is provided by adaptive sub-stepping, while the NN correction improves accuracy at close-encounter steps.

These two constraints are orthogonal: adaptive sub-stepping operates at the integration level (when to subdivide a step), while analytic force direction operates at the force computation level (how to compute the force direction at each step). In the controlled ablations tested here, removing either adaptive close-encounter resolution or analytic direction enforcement leads to unbounded trajectories, while removing the NN correction does not.

The broader implication is that neural models for chaotic physical systems should not freely learn quantities whose symmetries are already known exactly. In this work, Newtonian radial direction is enforced analytically, and the neural network is restricted to a low-dimensional scalar correction during close encounters. This suggests a general design principle for hybrid scientific machine learning: preserve exact physical structure where available, and use learning only for targeted corrections where analytic or numerical approximations are most fragile. Although demonstrated here only for the unrestricted three-body problem, this principle may be relevant to larger N-body simulations and other dynamical systems where rare close interactions dominate long-horizon stability.

6.2 Positioning SIMON Relative to `ias15`

`SIMON` does not compete with `ias15` on energy conservation accuracy. `ias15` uses continuous global adaptive timestepping at every step, achieving float64-precision energy conservation ($\approx 10^{-14}\%$). `SIMON` uses a fixed leapfrog timestep with targeted sub-stepping at close encounters, achieving 0.67–13.77% energy drift at $dt = 0.04\,\mathrm{yr}$—orders of magnitude larger, but operationally acceptable for the targeted use case.

The relevant comparison is not accuracy but *operating point*: `SIMON` targets long-horizon orbital characterisation at 1.12× the speed of `ias15`. In a chaotic three-body system with $\lambda \approx 0.17\,\mathrm{yr}^{-1}$, the Lyapunov time is $1/\lambda \approx 6\,\mathrm{yr}$. Beyond this timescale, exact trajectory reproduction is impossible regardless of integrator: both `SIMON` and `ias15` diverge exponentially from the "true" trajectory. The appropriate question is not "is `SIMON` as accurate as `ias15`" but "does `SIMON` reproduce the correct qualitative dynamical behaviour—bounded vs. unbounded, chaotic divergence rate, physical instability—at reduced computational cost?" The results confirm that it does.

These are the primary conclusions of this work:

- **`SIMON` identifies a stability-preserving operating point:** The optimised hybrid model achieves a 1.12× speedup over `Rebound ias15` at the operational timestep $dt = 0.04\,\mathrm{yr}$, while preserving bounded dynamics across all tested stable configurations. A larger 2.18× speedup is observed at the coarser timestep $dt = 0.08\,\mathrm{yr}$, but this point is best interpreted as part of the speed–accuracy frontier rather than the main operating regime.
- **Adaptive sub-stepping provides the primary stability guarantee:** The introduction of distance-based dt adaptation during close encounters allows `SIMON` to preserve bounded orbital dynamics for all three bodies across all tested stable initial conditions over the full 100-year horizon.
- **The dominant failure mode is phase drift:** All three bodies maintain qualitatively correct orbital geometry, tracking the bounded orbital structure of `ias15`. The primary disagreement is a gradual timing mismatch that accumulates after 25–

30 years, consistent with the chaotic divergence inherent to the Three-Body Problem (see Section 2.3).

- **Divergence rate:** `SIMON` produces a divergence-rate proxy $\lambda \approx 0.1655\,\mathrm{yr}^{-1}$ (10–50 yr fit window), consistent with the chaotic nature of the Three-Body Problem. The positive value confirms chaotic sensitivity. The exponential growth with no discontinuities demonstrates that `SIMON` reproduces the system's chaotic divergence rate without introducing catastrophic instabilities over the tested horizon.
- **Discovered what controls runtime:** Cost is dominated by force evaluation + gating + framework overhead, not integrator arithmetic, and so to improve speed, optimising data movement is more important than tuning dt.
- **The neural network correction is an accuracy mechanism, not a stability mechanism:** The ablation study (Run C) demonstrates that removing the NN correction while retaining adaptive sub-stepping produces bounded dynamics with 8.5% larger final RMS. Adaptive sub-stepping is the primary stability mechanism; the NN provides a targeted accuracy refinement at close-encounter steps.
- **Developed reusable methodology for future analysis:** The evaluation statistics provide quantitative and qualitative insights into numerical stability, speed, and accuracy for chaotic systems. Training reproducibility is confirmed across five random seeds.

We found that hybrid learned force corrections can preserve qualitative three-body dynamics over 100-year horizons while accumulating phase drift, and that long-horizon performance is best summarised via a divergence-rate proxy rather than point-wise trajectory error. Computational performance is best interpreted through a speed–accuracy frontier: at the operational timestep, `SIMON` provides a modest cost reduction while preserving bounded dynamics, whereas coarser timesteps illustrate the trade-off between speed and stability rather than defining the main operating regime.

To answer the research question:

> *How closely can physically constrained neural network force models approximate the long-term stability of traditional numerical integrators in three-body simulations, while reducing computational cost without sacrificing qualitative dynamical stability?*

A physically constrained neural network can approximate the qualitative long-horizon behaviour of traditional numerical integrators in the tested three-body regimes. All configurations that remain bounded under `ias15` also remain bounded under `SIMON`, and the configurations that eject under `ias15` are also identified as physically unstable by `SIMON`. At the operational timestep, this agreement is achieved with a modest reduction in computational cost rather than a large raw speedup. The main conclusion is therefore not that SIMON replaces ias15, but that neural force correction can be made dynamically usable in chaotic gravitational simulation when exact physical structure is enforced analytically.

Stability is preserved through two mechanisms: a gating system that falls back to exact Newtonian gravity when the correction factor $c \notin [0.2, 5]$, and an adaptive sub-stepping mechanism that adjusts the timestep according to the gravitational acceleration during close encounters.

On computational speed, the observed advantage arises from three separable effects. First, SIMON uses a fixed macro-timestep rather than ias15's continuously adaptive high-order stepping, reducing the number of integration steps at coarser timesteps. Second, the NN correction is invoked only during close encounters, so its runtime overhead is limited. Third, adaptive sub-stepping adds overhead during close encounters but is necessary for stability. Thus, the speedup is not caused by faster NN force evaluation alone; it is the net result of timestep choice, targeted NN invocation, and close-encounter sub-stepping.

6.3 Limitations and Scope

Several limitations should be noted.

Single initial condition for most experiments. The ablation study (Section 5.1) uses IC1 exclusively. Results may differ for initial conditions with different mass ratios or orbital geometries, although the multi-IC evaluation (Section 5.2) provides partial evidence for generalisation.

λ as a proxy. The fitted divergence-rate λ reflects both the intrinsic chaotic sensitivity of the three-body system and SIMON's accumulated force errors. It is not a formal Lyapunov exponent: we compare two different models rather than two nearby initial conditions evolved under the same dynamics [Bishop, 2025].

Fixed operational timestep. The $dt = 0.04\,\mathrm{yr}$ operational point was selected from the speed-accuracy frontier for this specific system. A different system or mass ratio may require a different dt.

High-eccentricity instability. High-eccentricity configurations with the mass ratios tested proved intrinsically dynamically unstable in both attempts (IC5 and IC7), confirmed independently by ias15. It is not known whether a bounded high-eccentricity case exists for these mass ratios.

This work should therefore be interpreted as a controlled methodological study rather than as a universal replacement for high-order reference integrators such as ias15. The evidence supports the narrower claim that, for the tested three-body regimes, physically constrained neural correction can preserve qualitative bounded–unbounded behaviour when exact radial direction and adaptive close-encounter resolution are enforced. It does not claim universal superiority over `ias15`, nor does it claim that the same thresholds, timestep, or neural architecture will transfer unchanged to arbitrary N-body systems.

6.4 Future Work

Several extensions are natural:

- **Extension to $N > 3$ bodies.** The pairwise force architecture scales naturally to larger N; evaluating performance on four- and five-body systems is a direct next step.
- **Variable mass ratios.** The current training distribution is fixed at $[1.0, 0.01, 0.005]\, M_\odot$. Training on a broader mass-ratio distribution would improve generalisation.
- **Longer time horizons.** Evaluating SIMON over $T = 1000\,\mathrm{yr}$ would test whether phase drift remains bounded or diverges structurally.
- **Comparison with equivariant neural force models.** Architectures that enforce rotational and translational equivariance by construction (e.g. graph neural networks with spherical harmonic features) may provide stronger stability guarantees than the scalar correction approach.

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